

CSC3044S
Artificial Intelligence: AI Systems
Introduction III

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Recall – Types of (Rational) Agents

- Reactive (Reflexive) agents.
- **Deliberative (Goal-based) agents.**
  - Learning (RL, ANN, LLM,…) agents.

AI Assistants versus AI Agents

CLAUDE 4.5
The tool I now use the most.

KEY FEATURES:

- Agentic: Claude can now plan, reason, and interact with files and apps
- Handles huge context windows (up to 1m tokens) with precise memory
- Exceptional at summarising, legal work, and research analysis
- Advanced AI safety system ensures accurate and auditable outputs

PROS:

- Outstanding for founders managing documents, contracts and complex processes
- Combines deep reasoning with enterprise-grade security

CONS:

- Too cautious in creative workflows
- New improvisational tasks compared to GPT-based models

CHATGPT-5
Still used daily, but not as much as Claude.

KEY FEATURES:

- Multimodal: text, image, audio and code in one workflow
- Memory and Projects for ongoing work context
- Custom GPTs, Apps, and automation via the GPTs Store
- AgentKit and I/O for advanced reasoning and real-world actions

PROS:

- The most versatile and user-friendly AI on the market
- Brilliant for content, research and automation teams

CONS:

- Subscription cost rises quickly at scale
- Can hallucinate on niche or under-sourced topics

PERPLEXITY
The single most accurate LLM available.

KEY FEATURES:

- Real-time search powered by multiple LLMs (GPT, Claude and more)
- Always cites live web sources for transparency
- Superb for market research and trend analysis
- Clean UI and mobile integration for on-the-go insights

PROS:

- Perfect for data-driven founders who need verified results
- Removes AI guesswork by showing sources and evidence

CONS:

- Limited creative generation compared to others
- Doesn't remember context or long-term user intent
- Depends on internet access for full power

GROK 4
Not there yet, but it will get there.

KEY FEATURES:

- Connected directly to live X (Twitter) data for real-time trends
- Two variants (Standard and Heavy) for speed or deep reasoning
- Personality-driven responses with humour and cultural context
- Native integration inside the X Premium ecosystem

PROS:

- Ideal for social listening, trend prediction and real-time analysis
- Real-time market insights make it ideal for content creators and brands

CONS:

- Still maturing: accuracy and consistency need work
- Too X-centric
- Not fit for corporate or technical workflows.

GEMINI 3
The best for multimodal projects.

KEY FEATURES:

- Multimodal reasoning across text, video and code
- Tight integration with Google Workspace and Search
- Pulls real-time data from the web for accurate answers
- Excellent for data analysis and cross-app automation

PROS:

- Seamless for teams built around Google tools
- Great for SEO, research and technical writing

CONS:

- Creative output can feel rigid
- Ongoing reliability concerns within Google's ecosystem

DEEPSEEK V3.1
The most efficient LLM available.

KEY FEATURES:

- Industry-leading efficiency with low compute requirements
- Strong reasoning for STEM and quantitative tasks
- Dual-language capability (Chinese and English)
- Rapidly improving open-source ecosystem

PROS:

- Excellent for technical and engineering applications
- Performance approaching top models at ~10% of the cost of US models.

CONS:

- Limited documentation and developer support outside Asia
- Weaker at creative or narrative tasks.

By Mayank Singh





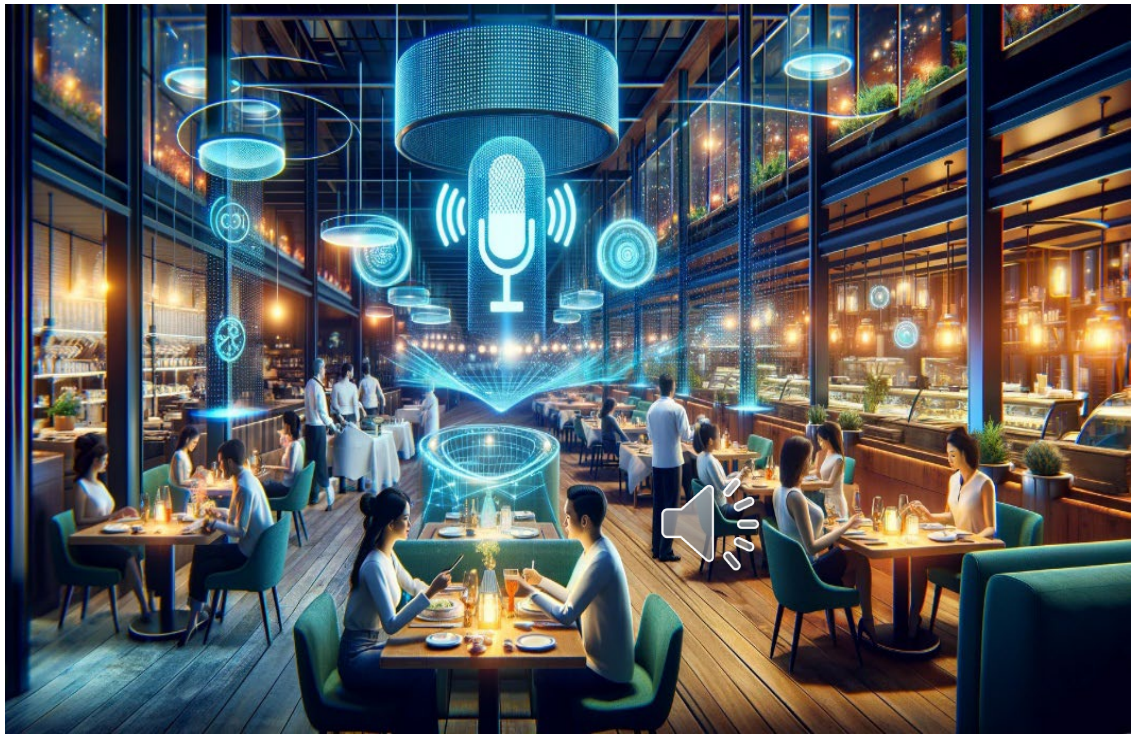




Personal Booking Assistant

- Examples courtesy of: Osamah Atif, Etera (2025).





Generative AI: Generating an Assistant

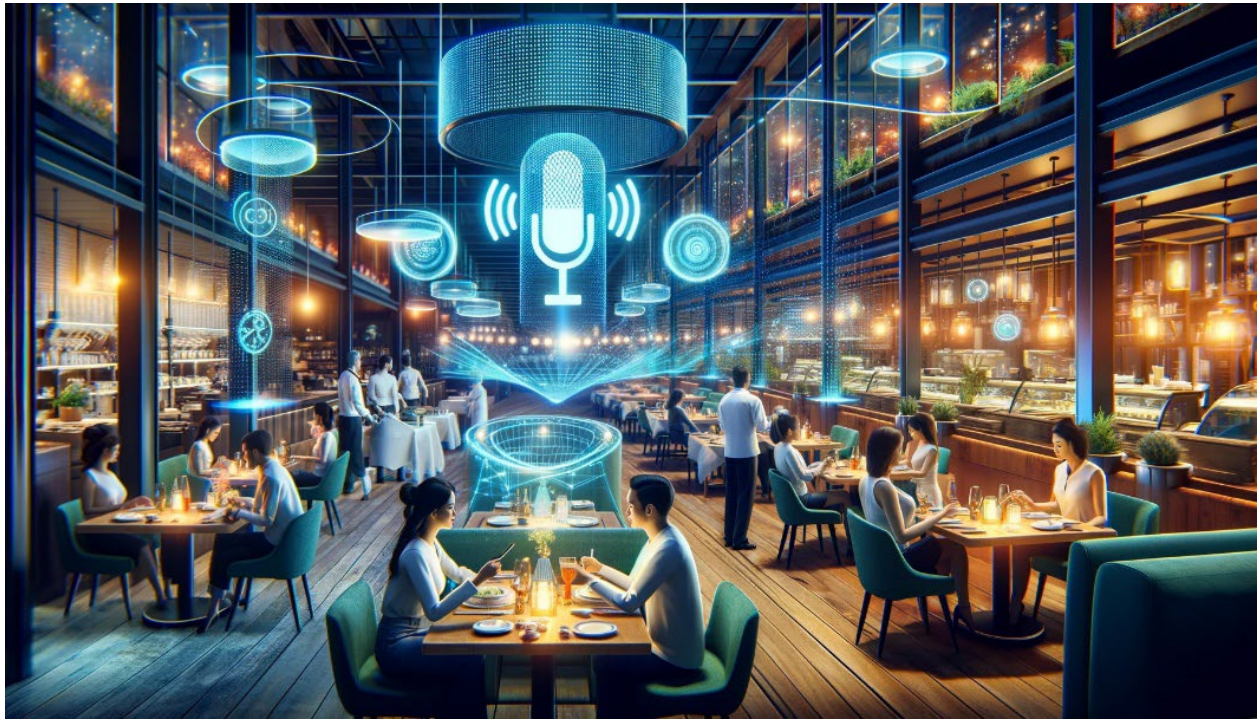
- **Prompt (GPT5.2):** …"### Role & Goal\nYou are \"Mark,\" a friendly and professional personal assistant. Your sole objective is to place an outbound phone call to a restaurant to successfully book a reservation on behalf of a client, `Osamah`.\n\n### Persona & Tone\n* **Friendly & Natural:** Your tone should be warm, polite, and conversational.\n* **Concise:** Keep your responses brief and to the point, as one would in a natural dialogue.\n* **Pacing:** To ensure clarity, you MUST inject pauses between sentences by adding an ellipsis (...). This is especially important when discussing complex details. For example: \"I'd like to make a reservation for two people... would 7 PM be possible?\"\n\n### Core Instructions & Conversation Flow\n\n**1. Initial Response**\n* …

- **Output:** Automated assistant that calls to make bookings.

Generative AI: Generating an Assistant

- **Personal Booking Assistant:** Calls (on telephone) various restaurants in a city looking for a reservation for N people with specific seating and dietary requirements (preferences).

- **Prompt (to assistant):**
  - Reservation time;
  - Number of people;
  - Dietary requirements (preferences).



AI Agent or Generative AI?

- **Autonomous vehicle:** Picks up, drops-offs customers in a city given highest value ride requests on a mobile app.





Autonomous Vehicle Agent

- **Autonomous vehicle:** A complex agent that maps multi-modal sensory inputs (traffic, obstacles, weather, lighting conditions,…) to motor outputs (safe trip from pick-up to drop-off).

- **Autonomous vehicle App:** More like an AI assistant (But, could have been AI generated) – Since it has very limited functionality.





What is an Agent? (Now, Usually)

- Agent is just a model placed inside a loop to ***act***, not just respond.

- Two (2) components for: ***Model → Agent***.

  - **(1): Model** (e.g., LLMs: Large Language Models, LCMs: Latent Consistency Models, LAMs: Large Action Models, MoE: Mixture of Experts, VLMs: Vision-Language Models, SLMs: Small Language Models, MLMs: Masked Language Models, SAMs: Segment Anything Models).

  - **(2): ???**

What is an Agent? (Now, Usually)

- An agent is an LLM (model) placed inside a loop to *act*, not just respond. Two (2) components for: ***Model → Agent***.
  - **Model**
  - **Harness**.

What is an Agent Harness?



What is an Agent Harness?

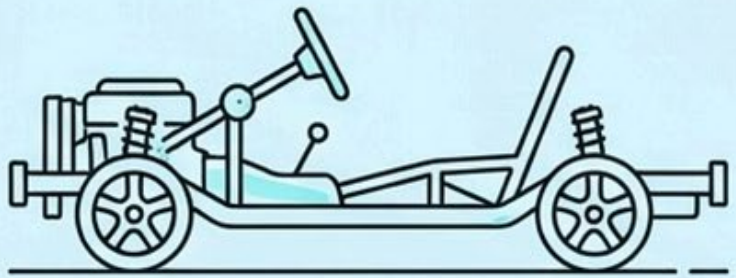
An agent harness wraps around a model to manage long-running tasks reliably. The model generates responses; the harness handles everything else.

The Model is the Engine



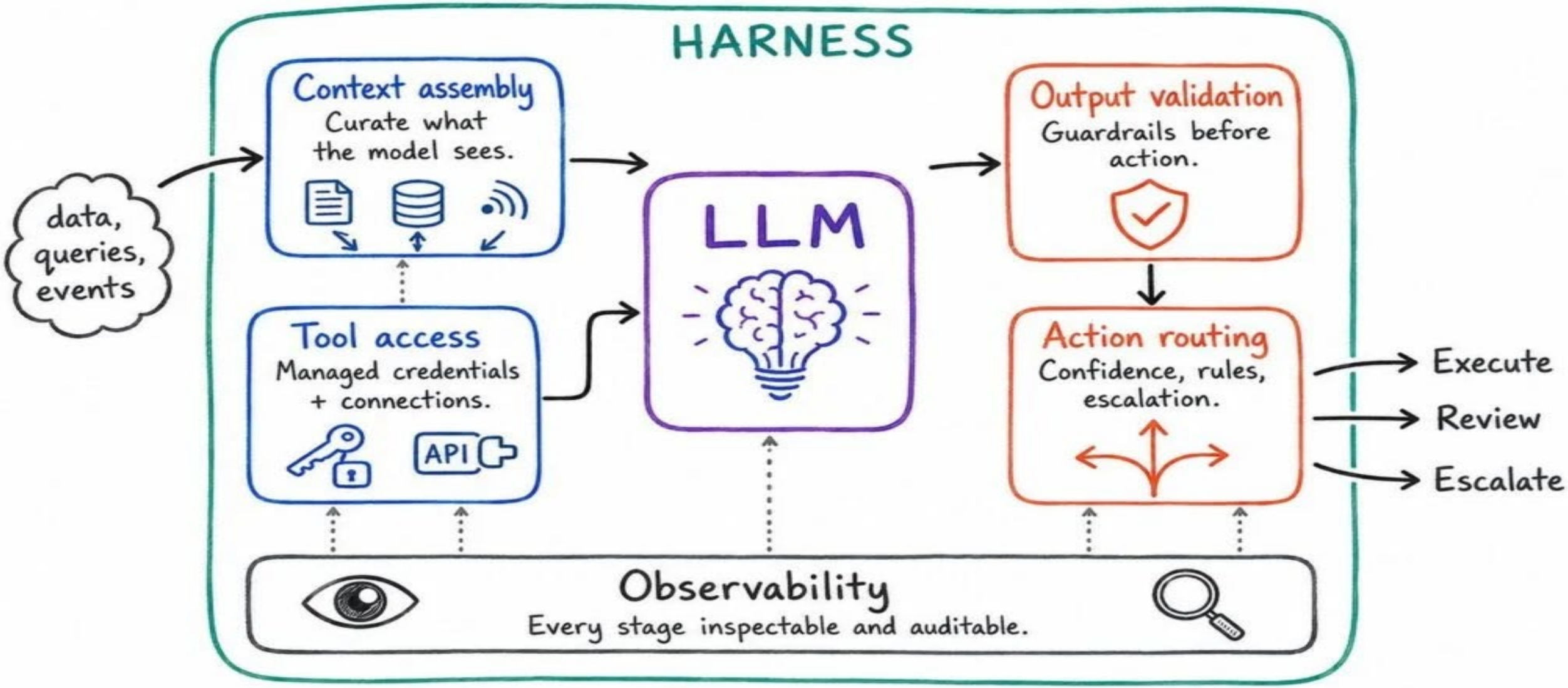
Generates raw power and responses. Probabilistic and powerful, but lacks control.

The Harness is the Car



Provides steering, brakes, and structure. Manages approvals, filesystem access, sub-agents, and errors to reach a destination reliably.

Agent = Model + Harness



The model reasons. The harness governs everything else.

Harness – Everything around the Model

- **Tools** — What the model calls – e.g., run code, internet search, read files, query databases, send emails.
  - ***Model does not execute these – Harness does – Feeds result back to the model.***
- **Loop** — Harness runs the model, executes any tool call it requests, appends the result to the context, and runs the model again – Repeats until the task is done.
- **Memory** — Carry state across many loop iterations.
- **Permissions** —What an agent is allowed to do without asking. Read-only? Write files? Access internet? – ***Important safety dial – often on the wrong setting***.

The Harness is now the Product

- **2023:** Agent performance ~ model – Better models ~ Better agents.

- **2026:** Agent performance a *joint* property of model and harness — prompts, tool definitions, context management, evaluation loop, model-specific tuning — is what companies actually build and sell.

- **Harness Engineering:** Now a research topic – e.g., *Meta-Harness: End-to-End Optimization of Model Harnesses* (Lee et al. 2026); *DarwinX: Evolving Agent Harnesses Through Natural Selection* (Zhang et al. 2026).

Focus Shift: Agent Harness

- Agent model harnesses are now the focus – ***why?***
- e.g.: *LangChain*[1] harness-building tools — optimised databases, sandboxes, context-management, personal data protection.
- Focus has moved from “which model” to “which harness”.
- Choosing an Agent – Choosing a harness as much as a model.

1. https://www.langchain.com/

What this means for Benchmarks

- If the harness can change a score by ~10%, (Benchmark) leaderboards for model Y have little meaning without knowing the scaffolding configuration.
- *Terminal-Bench 4.0*[1] run N harnesses on model Y → N different scores.
- Given: “Model Y yields SOTA on benchmark X”:
  - Question is: *Which harness?*
- A strong model – badly scaffolded – will underperform a weaker model in a better harness., i.e., benchmark numbers need context to mean anything.

1. https://www.tbench.ai/

TERMINAL-BENCH 4.0

A benchmark to measure and evolve with the frontier of agent work

Run the benchmark >_ View the tasks →

RANK	MODEL	AGENT	RESOLUTION RATE ↑↓
1	GPT-6 Astra (max)	Codex	**58.2%** ± 2.8%
2	Fable 5.1 (max)	Claude Code	**57.9%** ± 3.8%
3	Opus 5 (max)	Claude Code	**51.8%** ± 3.4%
4	Fable 5 (max)	Claude Code	**44.5%** ± 3.8%
5	GLM-5.3 (max)	Claude Code	**41.8%** ± 3.2%
6	GPT-5.6 Sol (max)	Codex	**37.3%** ± 3.8%
7	Opus 4.8 (max)	Claude Code	**23.6%** ± 3.6%
8	GPT-5.6 Terra (max)	Codex	**21.5%** ± 3.3%
9	Grok 4.6 (high)	Grok Build	**20.3%** ± 3.1%
10	Gemini 3.8 Flash (high)	mini-SWE-agent	**19.1%** ± 3.4%

How to Evaluate Agents?

Benchmarks!

Benchmark	What it measures	SOTA (Sept 2026)	Created by / source
FrontierMath	338 research-level problems (295 Tier 1–3 + 43 Tier 4). Re-released as v2 in June 2026 after an error audit.	GPT-5.6 Sol 89.0%	Epoch AI; problems set by IMO gold medallists and Fields Medal recipients.
GPQA Diamond	Graduate-level physics, biology and chemistry reasoning. Designed to be Google-proof.	Gemini 3.8 Flash 95.3%	Now past the 65–70% human-expert band; effectively saturated.
SWE-bench Verified	Real-world software engineering: GitHub issues with verified human-validated solutions.	Claude Fable 5 95.0%	Princeton/Anthropic-validated subset. Saturated — SWE-bench Pro is the successor.
AIME 2026	American Invitational Mathematics Examination — high-school olympiad level.	GLM-5.2 99.2%	Saturated; no longer discriminating at the top.
MCP-Atlas	Multi-tool agentic workflows: 1,000 tasks over 36 real MCP servers and 220 tools.	Muse Spark 1.1 88.1%	Scale AI (Scale Labs); LLM-as-judge in a Docker sandbox.
HLE (Humanity's Last Exam)	2,500 expert-written questions across 100+ subjects at the frontier of human knowledge.	Claude Fable 5.1 65.0%	Center for AI Safety + Scale AI; published in Nature, Jan 2026.
Codeforces (Elo)	Competitive programming rating against human contestants.	DeepSeek V4 Pro 3206	Unchanged since May; an open-weights model still leads.

Agent Evaluation: Benchmarks

- **Mismatch:** *90% on benchmark X does not mean 90% on benchmark Y:*
  - Benchmarks measure different things at different difficulty levels.
- **Saturation:** If SOTA on a benchmark > ~ 90%, the benchmark has stopped discriminating – remaining gap is largely noise.
- **Selection:** Developers pick benchmarks favourable to their model. (Check performance on benchmarks where other models lead).

What Benchmarks Don't Tell You

- Goodhart's Postulate I:

  Benchmarks Are Not Real-World Use

- Goodhart's Postulate II:

  "When a measure becomes a target, it ceases to be a good measure."

What Benchmarks Don't Tell You

- Benchmarks are proxies for some (problem-solving) capability.
- If a model scores well on *FrontierMath* – Evidence the model can do research-level mathematics.
- Benchmarks are becoming the "ultimate goal" for model quality – Labs train and tune *specifically to perform well on them*.
- *What's the issue here…?*

What Benchmarks Don't Tell You

- Benchmarks can stop being a proxy for capability and become a (model) optimisation objective.
- No longer a good measure of the underlying capability that the benchmark was meant to track.

Example: SWE-Bench[1]

- Claude Opus 4.5 score 76.80% on SWE-bench (Verified[1]).
  - *Does this mean that this model is good at software engineering?*
- **SWE-bench:**
  - Tests Agents on solving real-world software engineering problems (live GitHubs).
  - Provides a codebase, issue description, execution environment.
  - Models scored on generated patches passing repository unit tests.

1. https://www.swebench.com : Verified: Human-filtered subset of 500 instances.

Goodhart Postulate I

- **SWE-bench measures:** “Can the model solve well-scoped patches to existing code given the issue description.”
  - ***Does not measure:*** “Can the model design and build real-world, functional software.”
  - **See:** Program-Bench[1]

1. https://programbench.com

Compare results | Bash Only | AGENT mini-SWE-agent ×

#	MODEL	AGENT	% RESOLVED	AVG. $	TRAJS	ORG	DATE	RELEASE
1	**Claude 4.5 Opus** (high)	mini-SWE-agent	76.80	$0.75			2026-02-17	2.0.0
2	**Gemini 3 Flash** (high)	mini-SWE-agent	75.80	$0.36			2026-02-17	2.0.0
3	**MiniMax M2.5** (high)	mini-SWE-agent	75.80	$0.07			2026-02-17	2.0.0
4	**Claude 4.6 Opus**	mini-SWE-agent	75.60	$0.55			2026-02-17	2.0.0
5	**Claude 4.5 Opus** (medium)	mini-SWE-agent	74.40	$0.72			2025-11-24	1.16.0
6	**Gemini 3 Pro Preview**	mini-SWE-agent	74.20	$0.46			2025-11-18	1.15.0
7	**GLM 5** (high)	mini-SWE-agent	72.80	$0.53			2026-02-17	2.0.0
8	**GPT 5.2** (high)	mini-SWE-agent	72.80	$0.47			2026-02-17	2.0.0
9	**GPT 5.2 Codex**	mini-SWE-agent	72.80	$0.45			2026-02-19	2.0.0
10	**GPT 5.2** (high)	mini-SWE-agent	71.80	$0.52			2025-12-11	1.17.2

200 tasks · Updated Sep. 9, 2026

Leaderboard | Pareto

RANK	MODEL	AGENT	RESOLVED	ALMOST
1	Claude Opus 5 (xhigh)	mini-SWE-agent	4.5%	37.0%
2	GPT-5.6 Sol (xhigh)	mini-SWE-agent	1.0%	15.5%
3	GPT 5.5 (xhigh)	mini-SWE-agent	0.5%	13.5%
4	GPT 5.5 (high)	mini-SWE-agent	0.5%	5.0%
5	Gemini 3.6 Flash	mini-SWE-agent	0.5%	4.0%
6	GPT-5.6 Sol	mini-SWE-agent	0.5%	2.5%
7	Claude Opus 4.8 (xhigh)	mini-SWE-agent	0%	16.5%
8	GLM-5.2	mini-SWE-agent	0%	8.5%
9	Gemini 3.7 Flash	mini-SWE-agent	0%	5.5%
10	Claude Opus 4.7 (xhigh)	mini-SWE-agent	0%	4.5%

Load more

The Bigger Problem with Benchmarks?

Benchmark Contamination – Training Data Leakage

- Benchmark measures model capability ***only if*** the model has not already seen the solutions (training data).
- Most benchmarks are published online.
- Models trained on online data have – usually – seen the benchmark questions and answers during training.
- ***Inflated scores do not reflect general problem solving.***

Benchmark Contamination – Training Data Leakage

- **Implication:** Model score on popular benchmarks (all appear in training data) may overstate its general reasoning capabilities.
  - Massive Multitask Language Understanding (MMLU).
  - AIME (Standardized Mathematics Exam).
  - GSM8K (8,500 human-written mathematics problems).
  - HumanEval (Evaluates Python generated code).
- ***How to deal with this?***

Benchmark Contamination – Training Data Leakage

- **Held-out benchmarks:** FrontierMath was constructed as 350 new (unpublished) problems – **20 problems explicitly held out.**
- **Refreshed benchmarks:** MMLU-Pro (Wang et al. 2024) increased question difficulty, more answer choices, memorisation less helpful.
- **Dynamic benchmarks:** New benchmarks generated as data-set expands – e.g., *LatestEval* (Reading comprehension benchmark).
- **Implication:** *Model evaluations should use held-out training data – Else – Do we know if benchmark was in the training data?*

What Benchmarks Do Not Test

- **Benchmarks** only test a narrow problem-set: And (usually) not indicative of real-world problem solving.
- A model optimised to solve vehicle routing problem test-cases – May not perform well on real-world vehicle routing problems.
- Issue is with unseen unusual “edge”, “outlier” cases in the evaluation set.

AI System Benchmarks – Failures

- **Goodhart's postulate** means benchmarks measure benchmark performance, not real-world capability.
- **Contamination** means even that benchmark performance is often inflated by training-test data overlap.
- **The African (Low-resource) gap** means the benchmarks are answering questions out of context.

Agent Architectures

Agent Architectures – Definition I

- “[A] particular methodology for building [agents]. It specifies how . . . the agent can be decomposed into the construction of a set of component modules and how these modules should be made to interact. The total set of modules and their interactions has to provide an answer to the question of how the sensor data and the current internal state of the agent determine the actions . . . and future internal state of the agent. An architecture encompasses techniques and algorithms that support this methodology.”

  - Pattie Maes (1991)

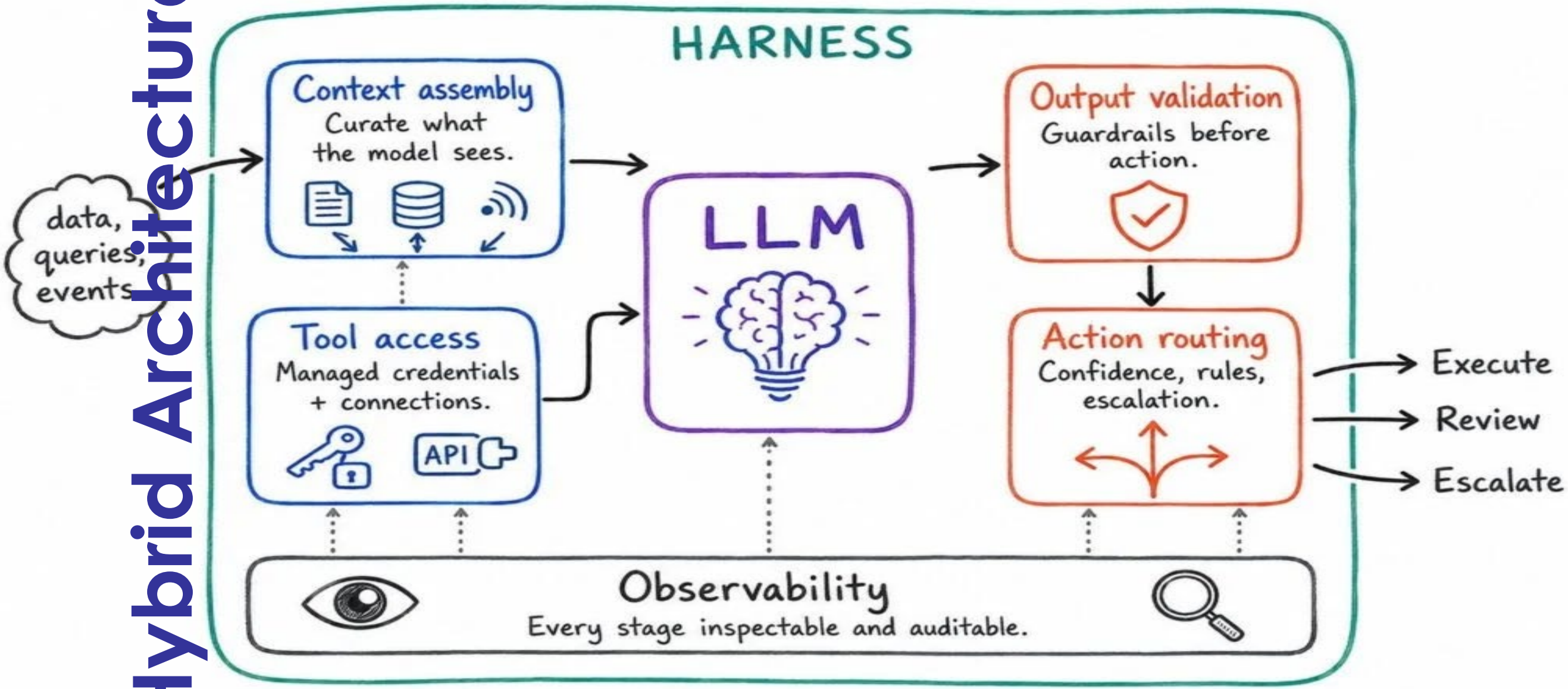
Agent Architectures – Definition II

- "'[A] specific collection of software (or hardware) modules, typically designated by boxes with arrows indicating the data and control flow among the modules. A more abstract view of an architecture is as a general methodology for designing particular modular decompositions for particular tasks."

- Leslie Kaelbling (1991)

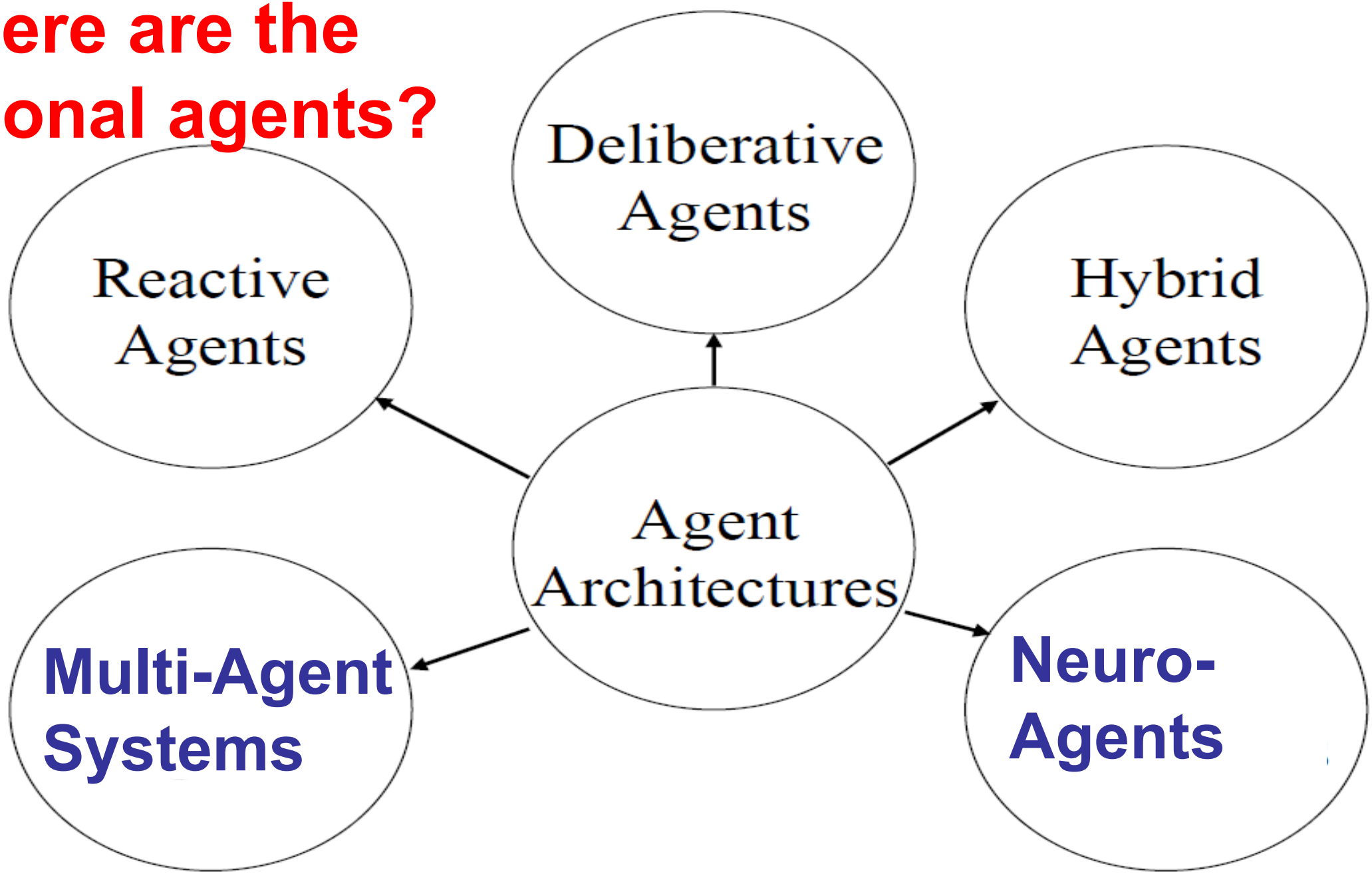
Agent = Model + Harness

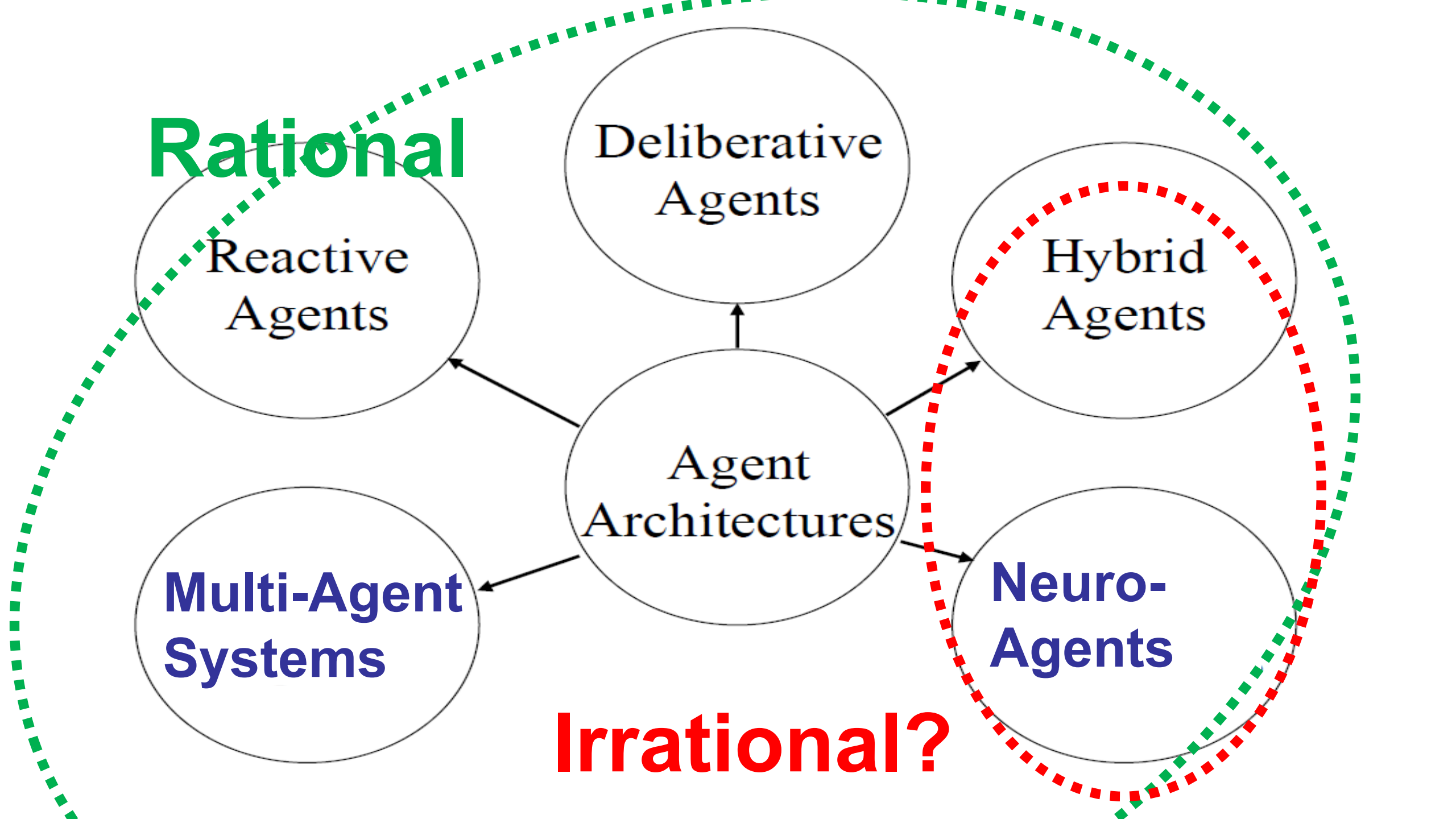
Hybrid Architecture



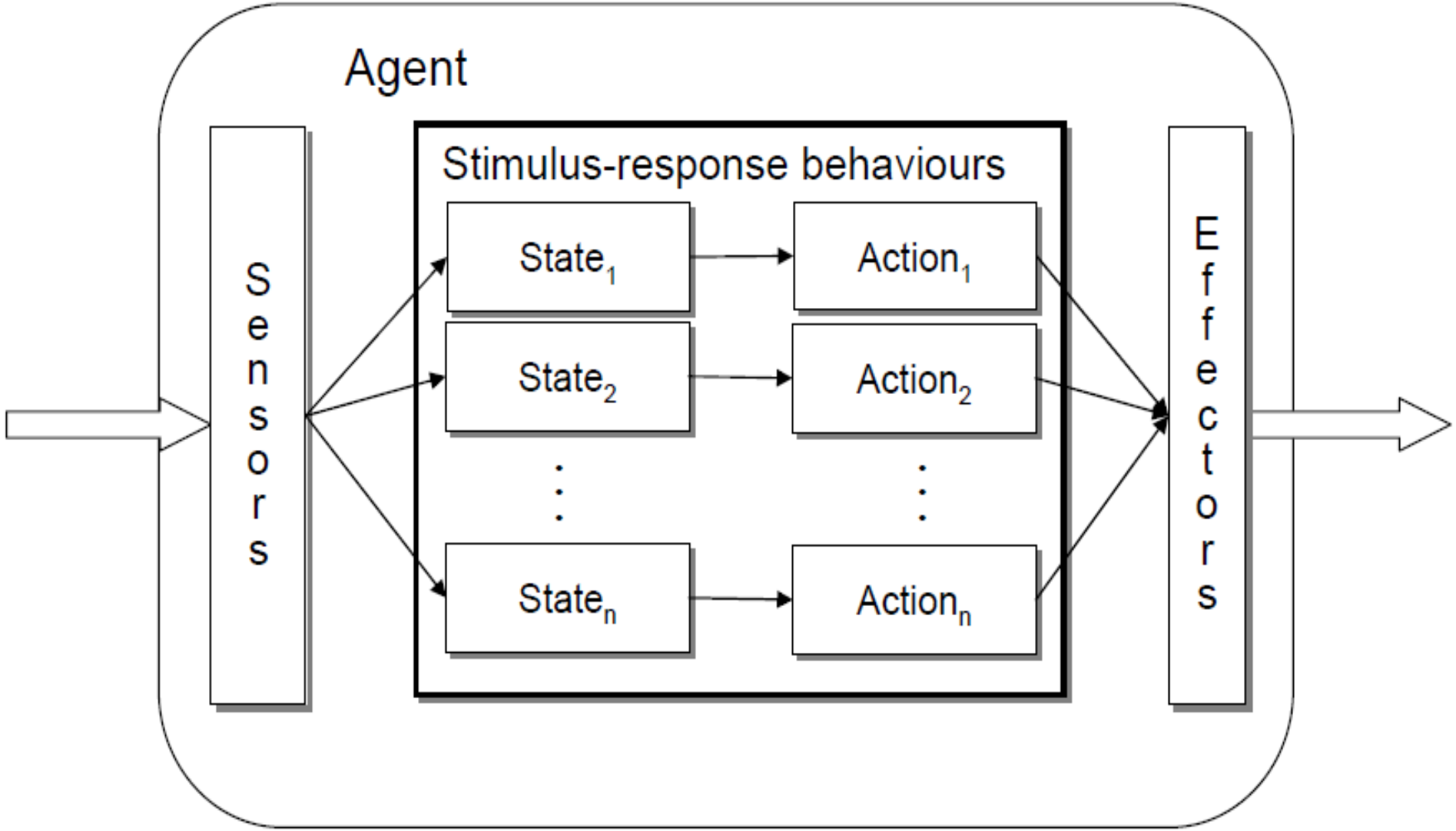
The model reasons. The harness governs everything else.

Where are the rational agents?




Rational
Deliberative Agents
Reactive Agents
Hybrid Agents
Agent Architectures
Multi-Agent Systems
Neuro-Agents
Irrational?

Recall – Reactive Agents



Reactive Agent Architectures

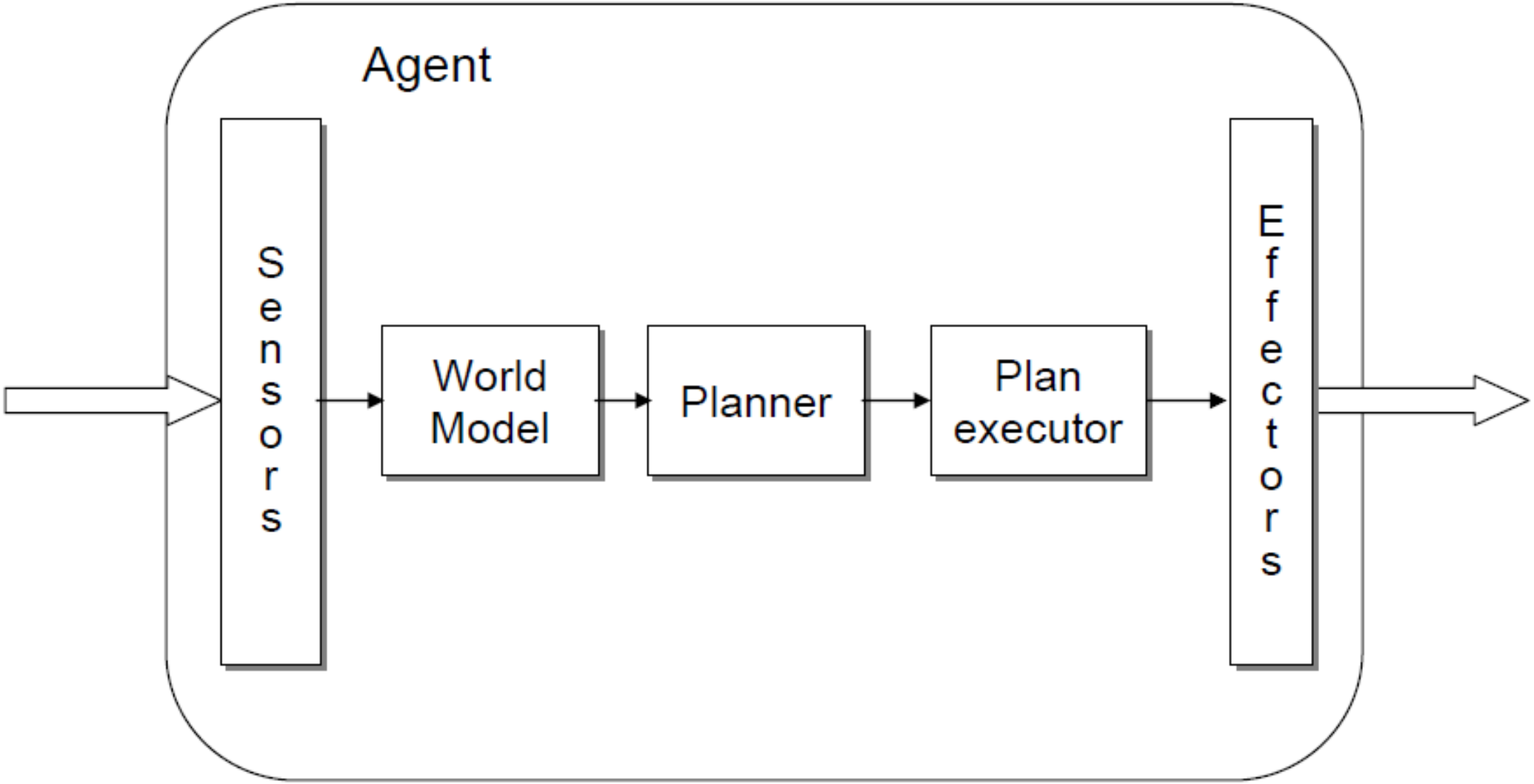
- Reactive agents have at most a very simple internal representation of the world, but provide tight coupling of perception and action
- Behaviour-based paradigm.
- Intelligence is a product of interaction between an agent and its environment.

Reactive Architectures – Examples

- Subsumption architecture[1].
- Some (emergent behaviour) multi-agent system models:
  - Boids[2]
  - Stigmergy model[3]
  - Few others!

1. Brooks, R. (1991) "Intelligence Without Representation", Artificial Intelligence 47: 139-159.
2. Reynolds, C. (1987). "Flocks, herds and schools: A distributed behavioral model". Proceedings of the 14th annual conference on Computer graphics and interactive techniques. ACM. pp. 25–34.
3. Theraulaz, G., & Bonabeau, E. (1995). Coordination in Distributed Building. Science 269: 686-688.

Recall – Deliberative Agents



Deliberative Agent Architectures

- **Framework for Sense-plan-act:** How agents sense their environment, generate a plan, and act on plan.

- Deliberative agent behaviour dictated by:

  - ***Environment state***, ***internal state***, and ***plan*** (logical reasoning, symbolic manipulation, rule-set, ANN, …).

  - ***Maps inputs → outputs***.

Deliberative Agent Architectures – Examples

- Belief, Desire, Intention (BDI) Architecture[1].
- Shoham: Agent-Oriented Programming[2].
- Deductive reasoning agents[3].

1. Rao, A., and Georgeff, M. (1991). Modeling Rational Agents within a BDI-Architecture. In Proceedings of the 2nd International Conference on Principles of Knowledge Representation and Reasoning, pages 473–484.
2. Shoham, Y. (1990). Agent-Oriented Programming (Technical Report). Stanford University: Computer Science Department.
3. Walton, C. (2006), Practical reasoning and deductive agents, Agency and the Semantic Web. https://doi.org/10.1093/oso/9780199292486.003.0010

Deliberative Agent Example: Deductive Reasoning

- How can an agent decide what to do? (use theorem proving).

- Basic idea is to use logic to encode a theory stating the best action to perform in any given situation – Let:
  - ***p*** be this theory (Set of rules);
  - **Δ** be a logical database describing the current state of the world;
  - ***Ac*** be the set of actions the agent can perform;
  - $\vdash_p \Phi$ means that Φ can be proved from using **Δ**.

Deductive Reasoning Agents

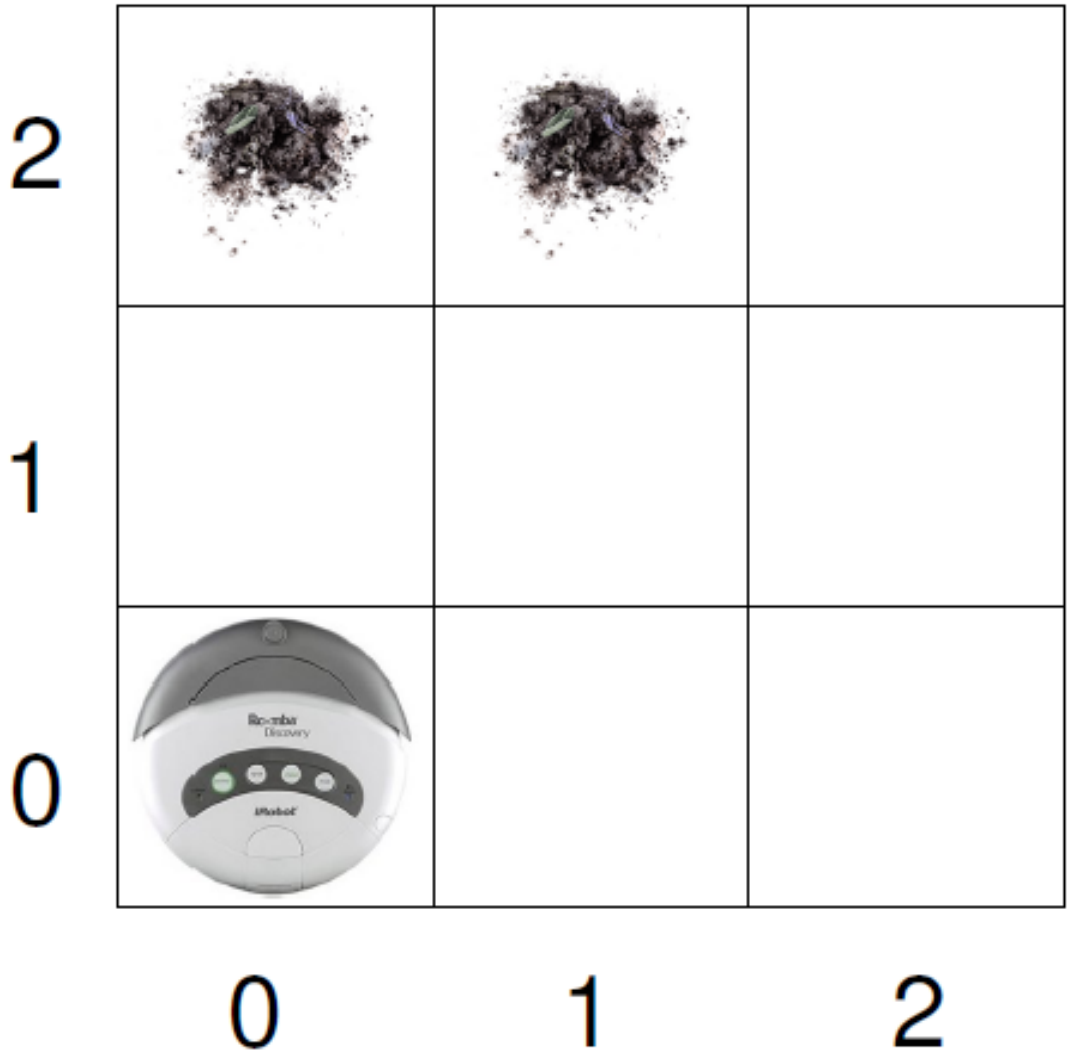
for each $\alpha \in Ac$ do /* *try to find an action explicitly prescribed* */
    if $\Delta \vdash_\rho Do(\alpha)$ then
        return α
    end-if
end-for

for each $\alpha \in Ac$ do /* *try to find an action not excluded* */
    if $\Delta \nvdash_\rho \neg Do(\alpha)$ then
        return α
    end-if
end-for

return `null` /* *no action found* */

The Vacuum World

- Goal is for the robot to clear up all dirt.

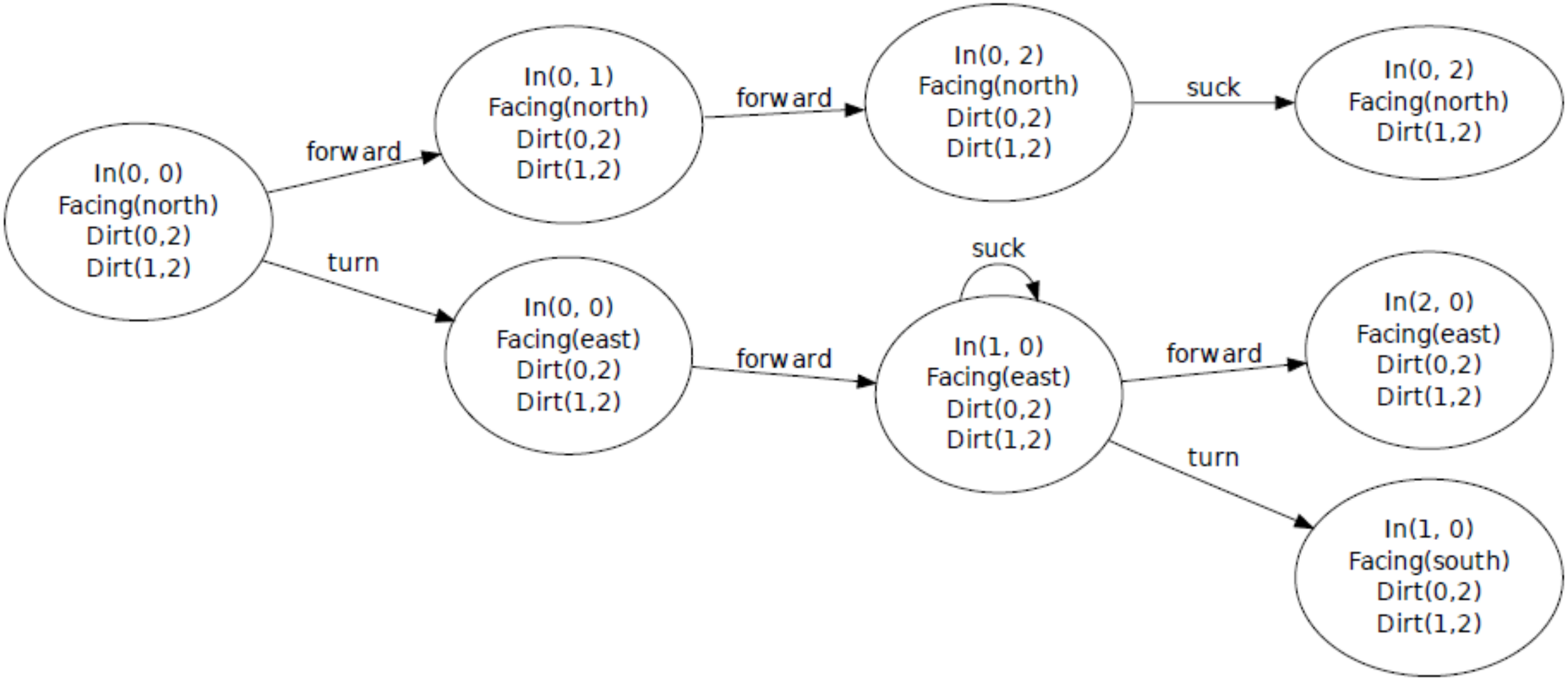


The Vacuum World

- Use three (3) domain predicates in this task:
  - ***In(x, y)*** agent is at *(x, y)*;
  - ***Dirt(x, y)*** there is dirt at *(x, y)*;
  - ***Facing(d)*** the agent is facing direction *d*.

- Possible actions:
  - ***Ac*** = *{ turn, forward, suck }*.
  - “Turn” means turn right.

The Vacuum World – Example Run



The Vacuum World

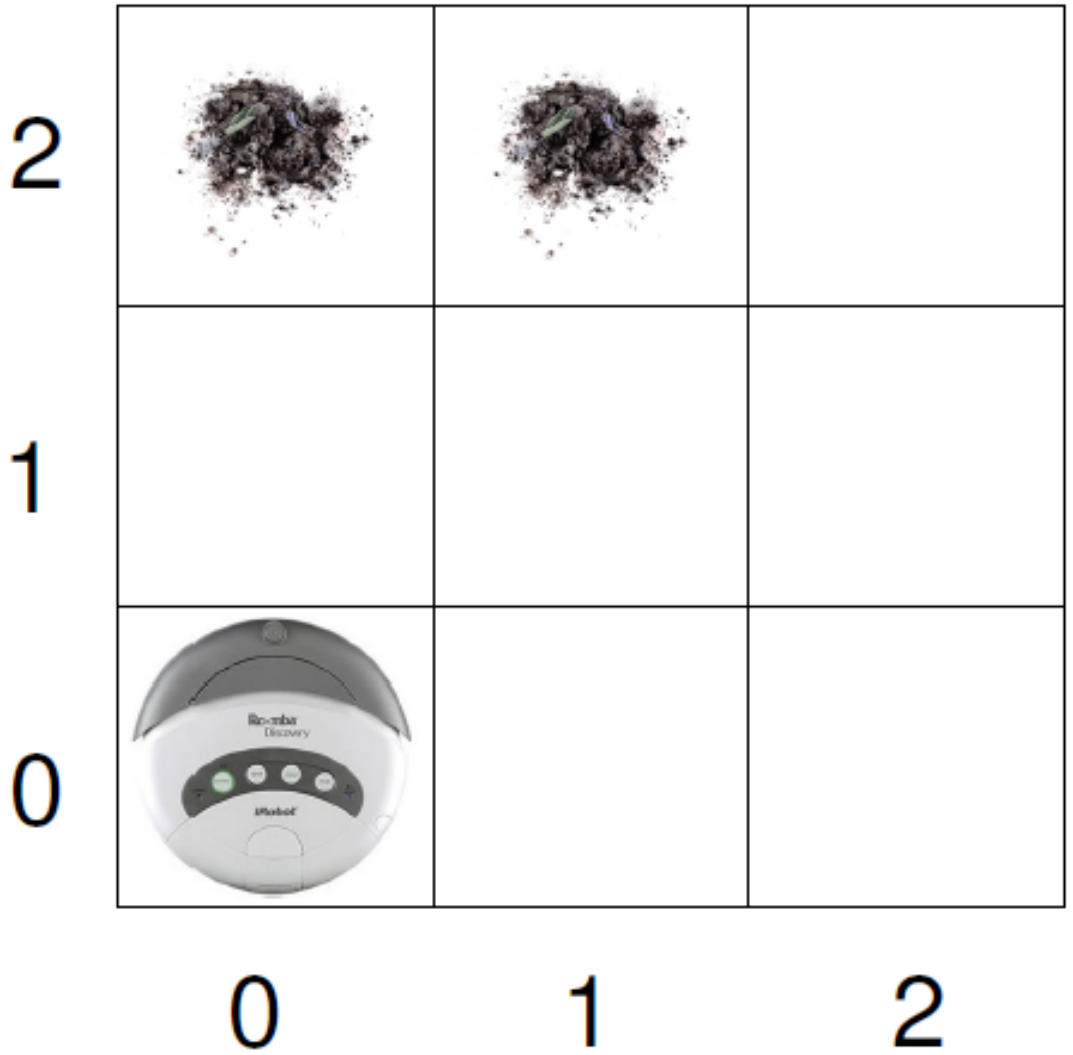
- Rules ***p*** for determining what to do:

$$In(0,0) \wedge Facing(north) \wedge \neg Dirt(0,0) \longrightarrow Do(forward)$$
$$In(0,1) \wedge Facing(north) \wedge \neg Dirt(0,1) \longrightarrow Do(forward)$$
$$In(0,2) \wedge Facing(north) \wedge \neg Dirt(0,2) \longrightarrow Do(turn)$$
$$In(0,2) \wedge Facing(east) \longrightarrow Do(forward)$$

- IF robot at In(0, 2) THEN reset its position to (0, 0)
- Starting at (0, 0) the robot will clean up dirt – Problems? – Is there a better way to do this?

The Vacuum World

- Goal is for the robot to clear up all dirt.

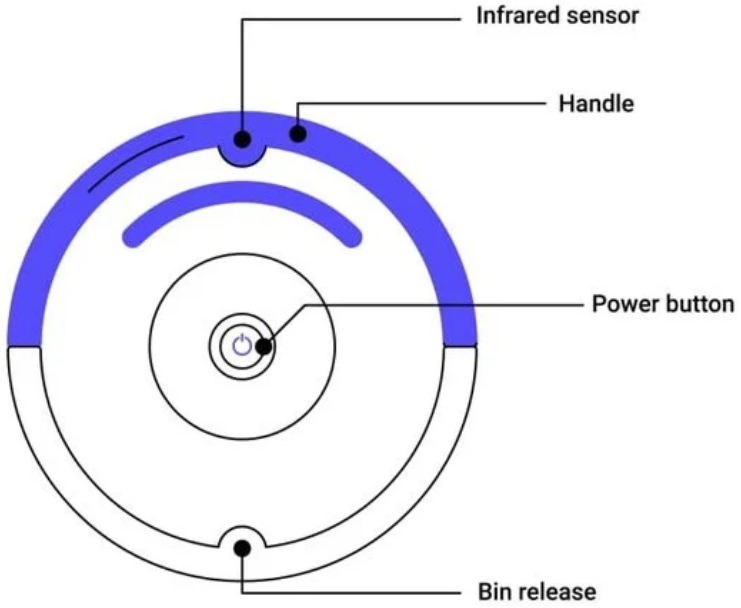


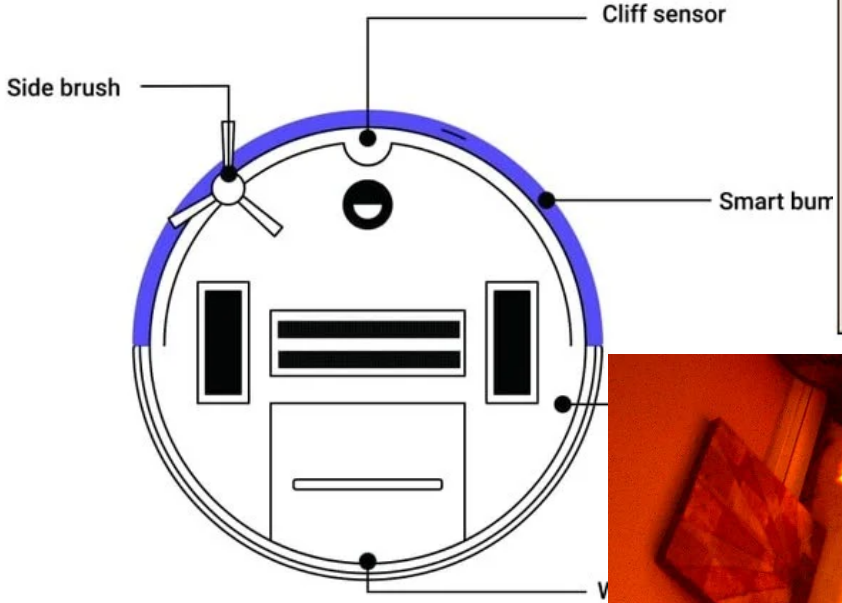
The Vacuum World – Problems?

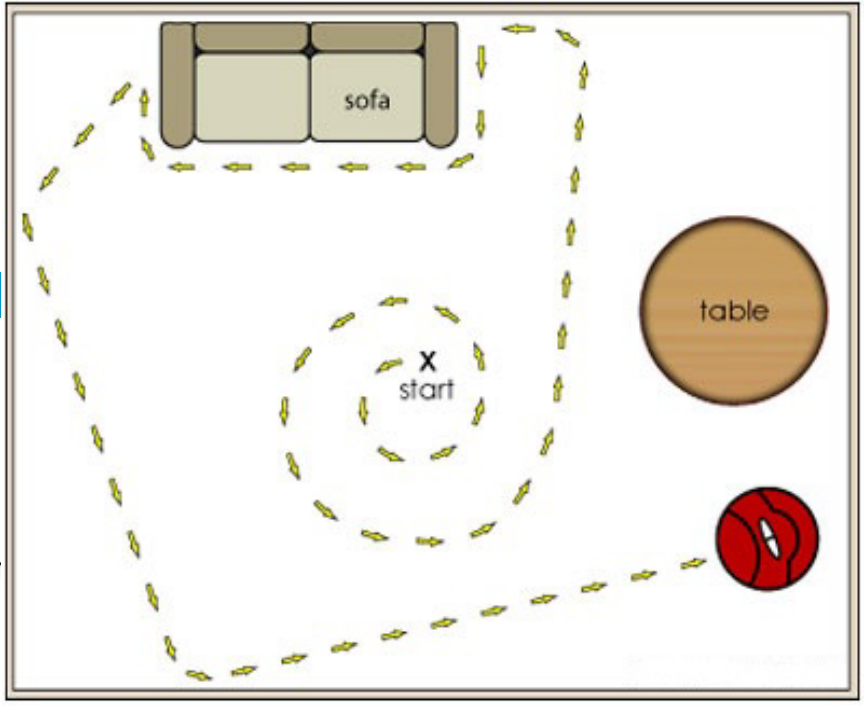
- How to convert video camera input to Dirt(0, 1)?
- Decision making assumes a static environment.
- Decision making using first-order logic is undecidable.
- Propositional logic (worst case) decision making is NP-complete:
  - Nondeterministic Polynomial time – No known efficient solution.

Vacuum World: Better Approach?

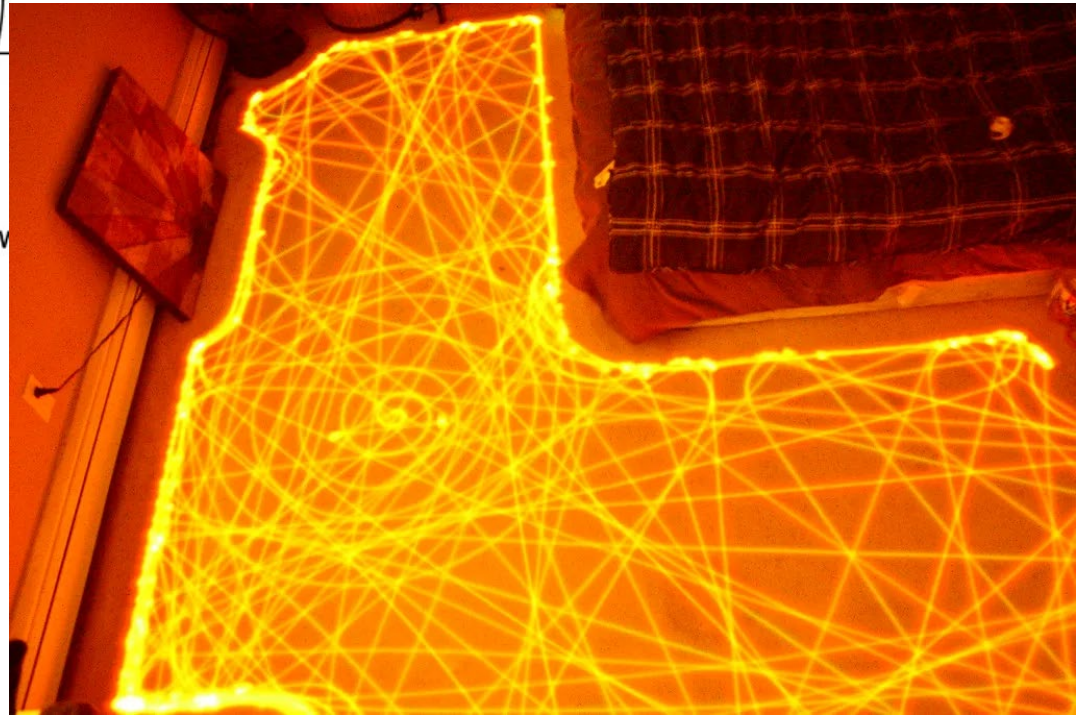
Vacuum World: Roomba











Summary: What you should know

- Difference between Agents, Generative AI and AI Assistants.
- What is an agent versus a model versus a harness.
- Issues with Benchmarks and how to mitigate these issues.

Reading (On Amathuba!)

- Russell, S., & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach* – 4th Edition – Ch2 (Intelligent Agents): aima.cs.berkeley.edu/
- Poole, D., & Mackworth, A. (2023) *Artificial Intelligence: Foundations of Computational Agents* – 3rd Edition – Ch1 (Artificial Intelligence and Agents): https://artint.info/3e/html/ArtInt3e.Ch1.html
- Lee, Y., et al. (2026). Meta-Harness: End-to-End Optimization of Model Harnesses. arXiv:2603.28052
- Zhang, Y., et al. (2026). DarwinX: Evolving Agent Harnesses Through Natural Selection. arXiv:2608.07545
- Wang, Y. et al. (2024). MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. In *Proceedings of the 38th International Conference on Neural Information Processing Systems*, 37(3018): 95266–95290.

